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Section : D

Global Feature Effect Exploration using built-in dataset.

Overview

This assignment studies **global feature effect explanation methods** in Explainable AI (XAI): **PDP, ICE, and ALE**.

Two black-box machine learning models are trained on the **Breast Cancer dataset (scikit-learn)** and their behaviour is analysed using these explanation techniques.

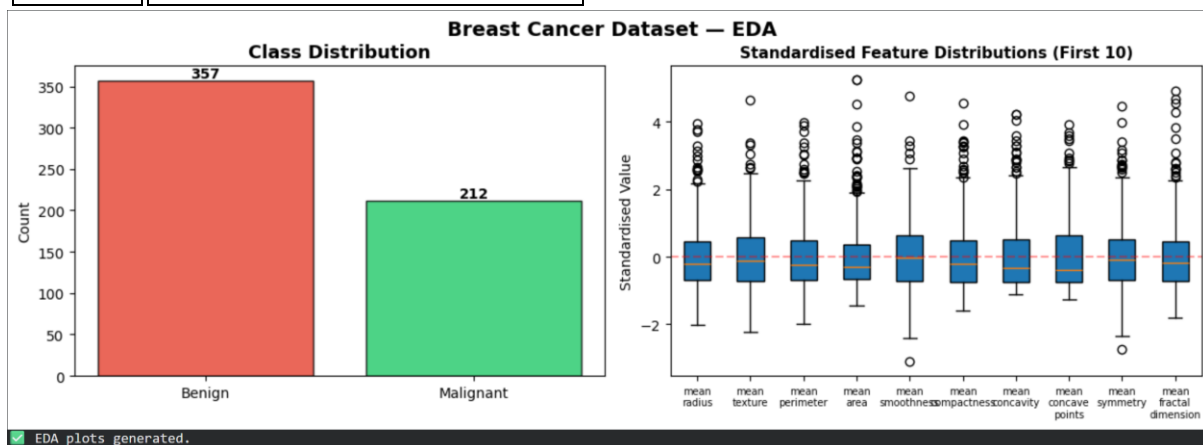
The goal is to understand:

- how features influence predictions
- how explanations differ between methods
- how explanations change across models

Dataset

Breast Cancer Wisconsin Dataset

Property	Value
Samples	569
Classes	212 Malignant, 357 Benign
Features	30 numerical features
Split	80% Train / 20% Test (Stratified)
Task	Binary Classification

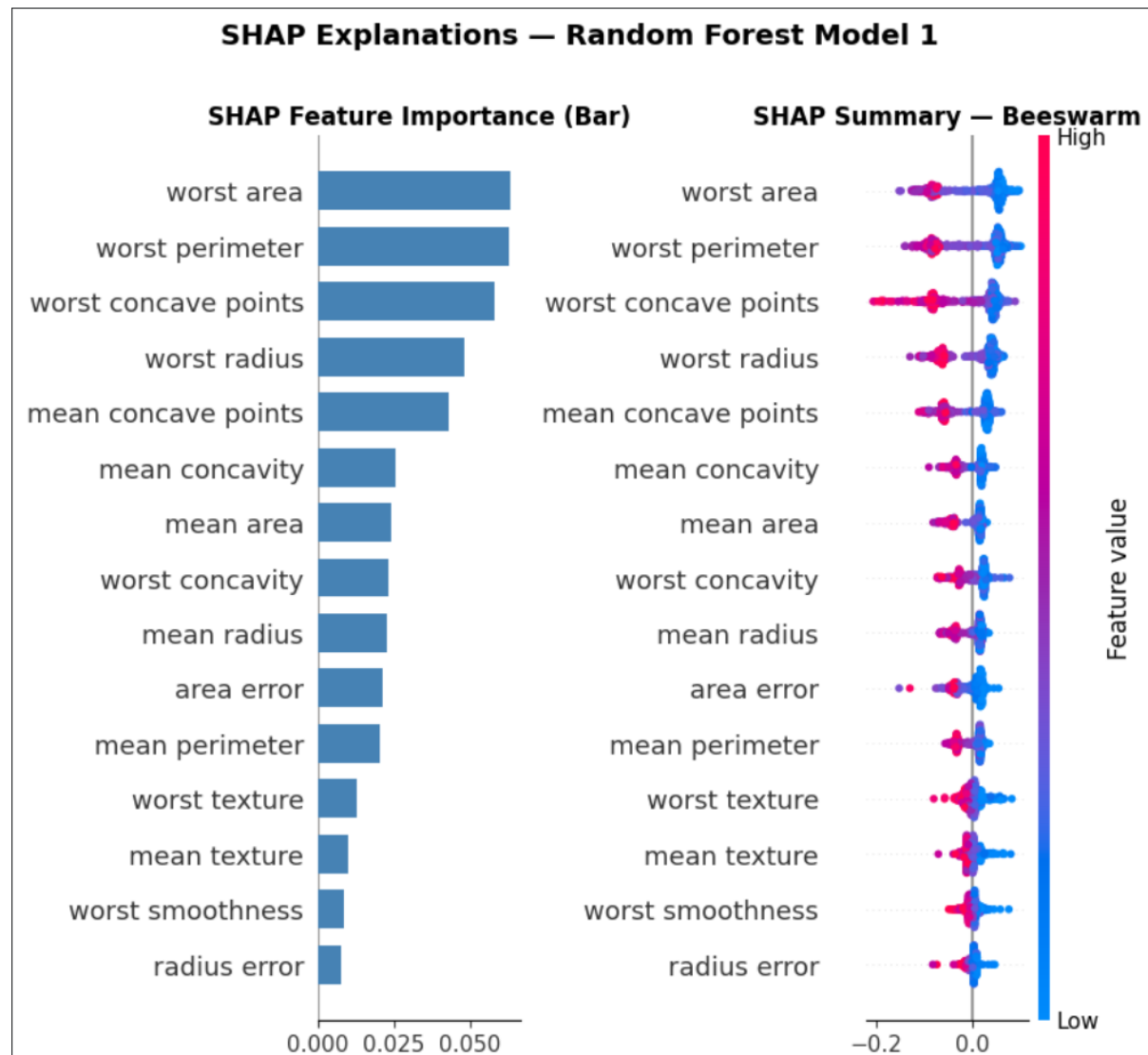


Models Used

Two models were trained on the same dataset:

Model 1 — Random Forest

- 200 trees
- max_depth = 10
- Test Accuracy $\approx$ 96–97%



Model 2 — Gradient Boosting Classifier

- 200 estimators
- learning_rate = 0.1
- max_depth = 4
- Test Accuracy $\approx$ 96–97%

Both models achieve similar accuracy, making them ideal for comparing explanations rather than performance.

Feature Selection (Using SHAP)

SHAP values ranked feature importance across all 30 features.

Selected features:

Feature	Importance
worst perimeter	High
worst concave points	High
worst texture	Moderate

SHAP identifies **which features matter**, while PDP/ICE/ALE explain **how they matter**.

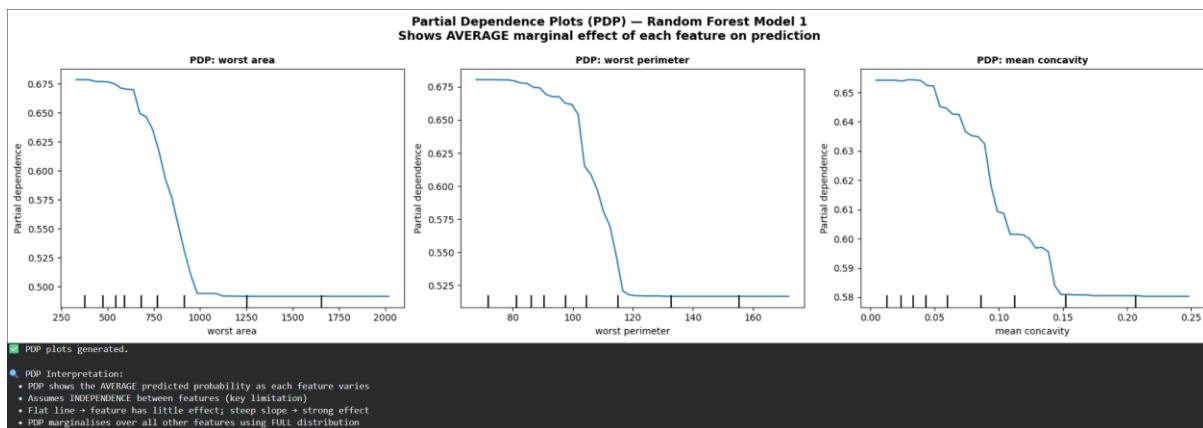
XAI Methods

1. PDP — Partial Dependence Plot

Shows the **average effect of a feature** on prediction by marginalising other features.

Limitation:

Assumes feature independence → unrealistic combinations may be evaluated.

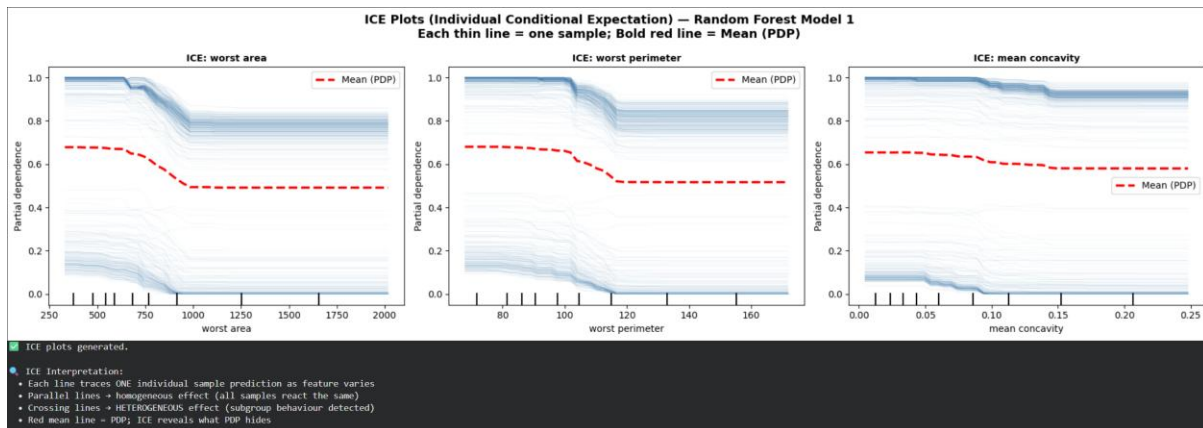


2. ICE — Individual Conditional Expectation

Plots prediction behaviour **per individual sample**.

Advantages:

- Reveals subgroup behaviour
- Mean of ICE curves = PDP

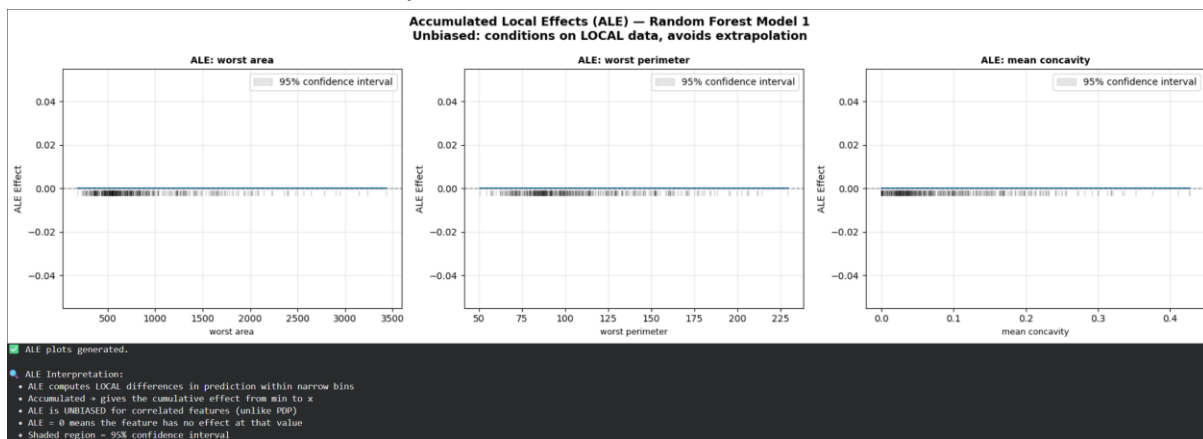


3. ALE — Accumulated Local Effects

Calculates feature effects using **local bins only**.

Advantages:

- Handles correlated features
- Avoids unrealistic data combinations
- Produces unbiased explanations



Experiments

Experiment A — Correlation Effect (PDP vs ALE)

Highly correlated feature pairs were analysed.

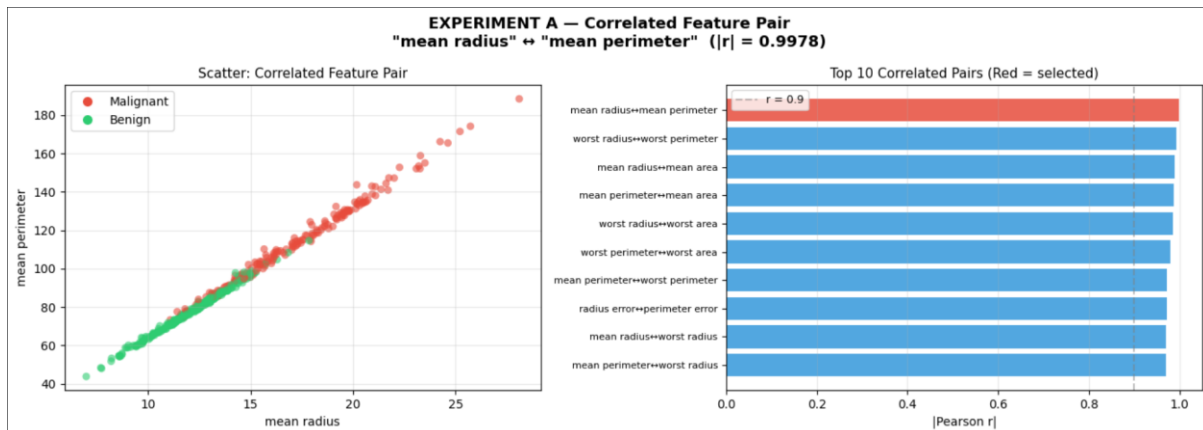
Observation

- PDP produced distorted/flat curves
- ALE produced sharper and realistic curves

Conclusion

PDP fails when features are correlated because it evaluates impossible feature combinations.

ALE gives trustworthy feature effects.



Experiment B — Subgroup Behaviour (ICE Analysis)

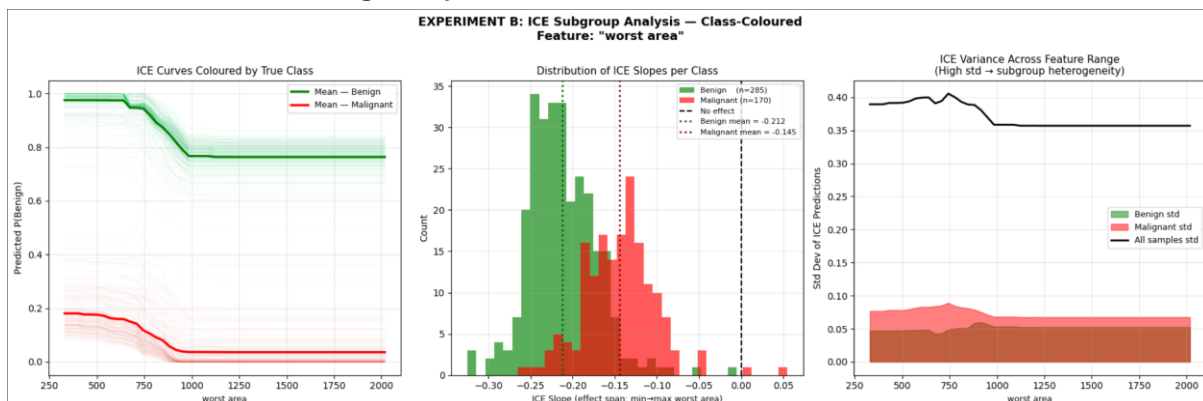
ICE curves plotted for *worst perimeter* and coloured by class.

Observation

- Malignant and Benign samples followed different patterns
- PDP hid these differences due to averaging
- Centred ICE highlighted subgroup behaviour clearly

Conclusion

ICE reveals hidden heterogeneity in data.



Experiment C — Model Difference (RF vs Gradient Boosting)

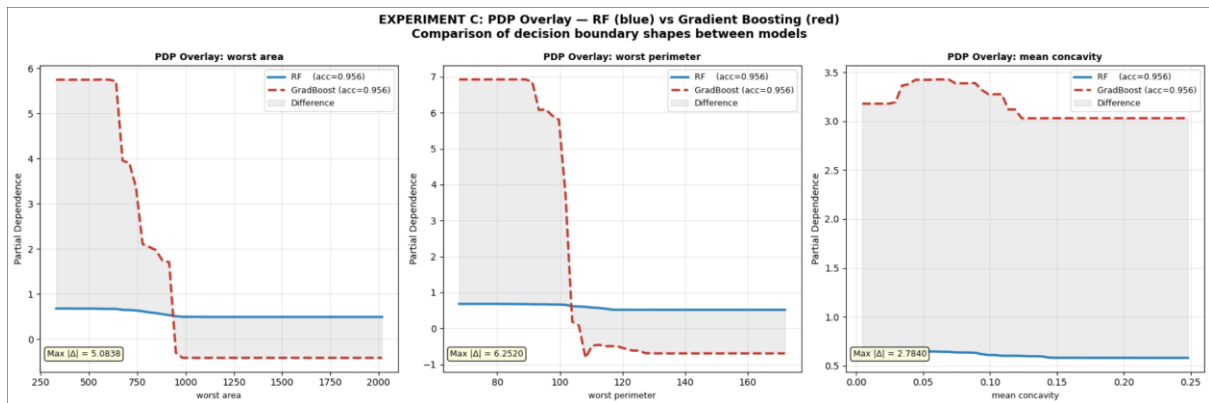
PDPs from both models were compared.

Observation

- Random Forest → smoother curves
- Gradient Boosting → sharper transitions

Conclusion

Explanations depend on the model, not just the dataset.



Comparison of Methods

Method	Handles Correlation	Shows Subgroups	Best Use
PDP	No	No	Quick global overview
ICE	No	Yes	Detect subgroup behaviour
ALE	Yes	No	Reliable effect for correlated features

Banana Problem — Complete Results Summary
Breast Cancer Dataset | PDP vs ICE vs ALE | 3 Experiments

